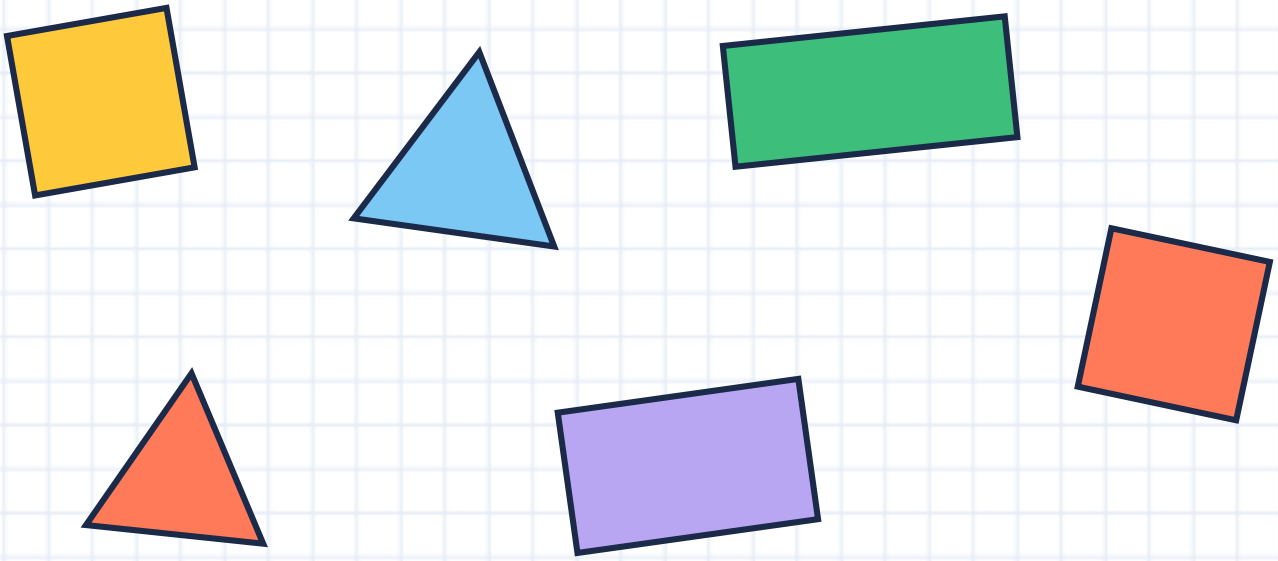


PRIMARY 2 · HOLIDAY PROGRAM

# Math Inquirers

## Notice it. Test it. Explain why.

**Name** _____ **Group** _____

### My Inquirer Rules

1. Notice the pattern.
2. Test it with objects first.
3. Check your answer.
4. Explain why.

# My Inquiry Journal

Your tutor stamps each lesson.

**Lesson 1**
**Digit Detectives**

**Lesson 2**
**Missing Digit Mysteries**

**Lesson 3**
**Legs, Wheels and Arrays**

**Lesson 4**
**Pattern Machines**

**Lesson 5**
**Mystery Number Machines**

**Lesson 6**
**Market Day**

**Lesson 7**
**Fair Shares**

**Lesson 8**
**Shape Builders**

**Lesson 9**
**Measure Masters**

**Lesson 10**
**Challenge Day**

Tutor: _____ Program dates: _____

**FAST TRACK.** Both Warm-up check answers right AND you can say why? Go to THINK HARDER and the Open investigation while the group does this page. Rejoin for check-together and talk time.

# DIGIT DETECTIVES

**Name** _____ **Group** _____ **Date** _____

A 3-digit number has hundreds, tens and ones. Where a digit sits tells you what it is worth.

## No.1

**Place value. Fill in the blanks.**

**a** 4 hundreds, 0 tens and 7 ones = _____

**b** 6 hundreds, 15 tens and 2 ones = _____

**c** 830 = 8 hundreds and _____ tens 830 = 7 hundreds and _____ tens

**d** In 572, the digit 7 is worth _____

## No.2

**Use the digit cards 2, 7 and 4. Use each card once to make a 3-digit number.**




**a** Make a list in order. Write every number you can make.

| Start with 2 | Start with 4 | Start with 7 |
|---|---|---|
| _____ _____ | _____ _____ | _____ _____ |

**b** Biggest number: _____ Smallest number: _____

**c** Which of your numbers is closest to 500? _____

**d** How many of your numbers are odd? _____

## No.3

**Now use the digit cards 0, 5 and 9.**




**a** List every 3-digit number you can make, in order.

**Numbers:** _____

**b** The cards 2, 7, 4 made 6 numbers. Why do 0, 5, 9 make fewer?

.....

.....

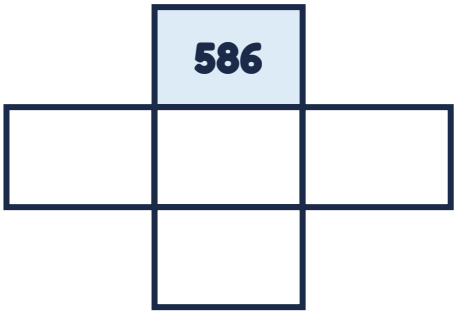
## No.4

**These are pieces cut from a 1000 chart. Each row has 10 numbers. Fill in the empty squares.**

**Piece A**

| | | |
|---|---|---|
| 347 | | |
| | 358 | |
| | | 369 |

**Piece B**




**a** On the chart, one square DOWN is _____ more. One square RIGHT is _____ more.

**b** Which number is 2 squares below 385 on the chart? _____

## No.5

**Look for the pattern. Fill in the next numbers.**

**a** 47, 147, 247, _____, _____

**b** 532, 522, 512, _____, _____

**c** 980, 985, 990, 995, _____

**d** 703, 702, 701, _____, _____

**No.6**

**Digit riddles. Who am I?**

**a** I am a 3-digit number. My hundreds digit is 4. My ones digit is 2 more than my tens digit. My digits add up to 12.

**I am:** _____

**b** I am a 3-digit number. My digits add up to 3. Find ALL the numbers I could be. Tip: use 3 counters. Put 1 counter in the hundreds box, then share the other 2 between the tens and the ones. Then try 2 counters in the hundreds box.

**How many numbers?** _____

# THINK HARDER

Name _____ Group _____ Date _____

*Draw a model, make a list or make a table. Then write how you know.*

## Stretch

**No.1**

**Use the digit cards 1, 5 and 8 once each. How many different 3-digit numbers can you make? List them in order.**

**No.2**

**Use the digit cards 4, 4 and 7. How many different 3-digit numbers can you make? Why is it not 6?**

**No.3**

**What number is 3 hundreds, 14 tens and 6 ones?**

**No.4**

**Mei says: "If I swap the tens digit and the ones digit of a 3-digit number, the new number is always smaller." Is she right? Show what happens with some examples. When does the number stay the same?**

## Challenge (for strong thinkers)

**No.5**

**Mei uses each of the digit cards 1, 2, 3, 4, 5 and 6 once to make two 3-digit numbers. She adds them. What is the smallest total she can make?**

**No.6**

**Ali counts in 5s, starting at 3: 3, 8, 13, 18 … Will he say 500? Will he say 998? Explain.**

**No.7**

**Ben writes all the numbers from 300 to 399. How many times does he write the digit 3?**

**No.8**

**Open investigation: Mirror numbers. A mirror number reads the same forwards and backwards, like 252 or 909. (a) Write 5 mirror numbers of your own. (b) Find ALL the mirror numbers from 200 to 299. How many are there? How do you know you have found them all? (c) Ali says every mirror number is odd. Is he right?**

**What was hard?** No marks for this page. Tick, then write one line.

This lesson felt: ☐ easy ☐ OK ☐ tricky

The hardest part was .....

I got unstuck by: ☐ made a list in order ☐ used digit cards or counters ☐ tried a smaller number first ☐ asked my partner

Next time I will try .....

**FAST TRACK.** Both Warm-up check answers right AND you can say why? Go to THINK HARDER and the Open investigation while the group does this page. Rejoin for check-together and talk time.

# MISSING DIGIT MYSTERIES

Name _____ Group _____ Date _____

Each yellow box hides one digit. Start at the ones column. Ask: was there a carry?

## No.1

**Find the missing digits. Write them in the boxes.**

**(a)**

$$\begin{array}{r} 2\square 4 \\ +\ 3\ 5\square \\ \hline 5\ 7\ 9 \end{array}$$

**(b)**

$$\begin{array}{r} 4\ 6\square \\ +\ \square\ 7\ 5 \\ \hline 8\ 3\ 7 \end{array}$$

**(c)**

$$\begin{array}{r} \square\ 8\ 6 \\ +\ 2\square 9 \\ \hline 6\ 4\ 5 \end{array}$$

Check one of your answers by adding the full numbers: _____ + _____ = _____

## No.2

**Find the missing digits. Try a digit and check it: the answer plus the number you take away must give the number you started with. Tip for (b): 3 – 7 cannot be done, so think 13 – 7.**

**(a)**

$$\begin{array}{r} 7\square 8 \\ -\ 3\ 2\square \\ \hline 4\ 5\ 3 \end{array}$$

**(b)**

$$\begin{array}{r} 5\square 3 \\ -\ \square\ 6\ 7 \\ \hline 2\ 7\ 6 \end{array}$$

## No.3

**The = sign means "is the same as". Fill in the boxes.**

**a** 350 + 140 = 300 + _____

**b** 625 – 200 = _____ – 100

**c** 480 + _____ = 500 + 260

**d** Ravi says 398 + 205 = 400 + 203 is true. Is he right? Explain how you know without adding.

## No.4

**Magic square. Every row, column and diagonal adds up to the same total.**

| | | |
|---|---|---|
| 80 | | 60 |
| | 50 | |
| 40 | | 20 |

**a** What is the magic total? (Find a line with all three numbers.) _____

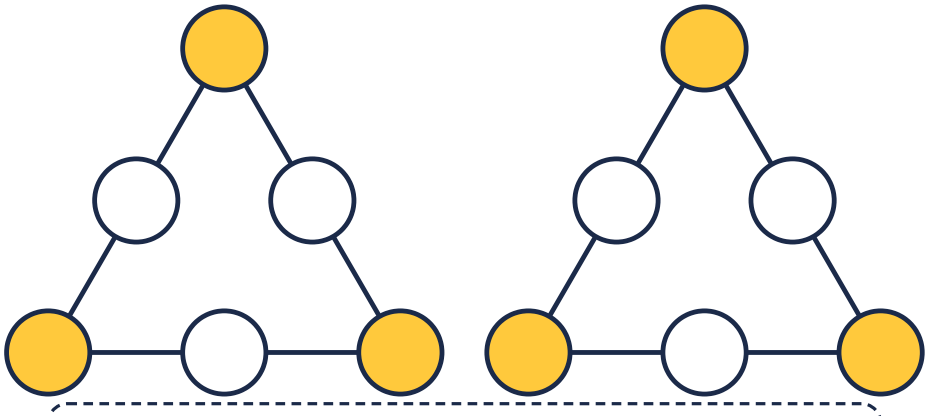
**b** Fill in the empty squares.

**c** Add 100 to every number. Is it still a magic square? What is the new total? Explain.

## No.5

**Magic triangles. Put the numbers 110, 120, 130, 140, 150 and 160 in the circles, once each, so every side has the same total. Do it for (a) and then (b).**

**(a) Each side = 390** **(b) Each side = 420**



**Number tiles: use all six in EACH triangle**

| 110 | 120 | 130 | 140 | 150 | 160 |
|---|---|---|---|---|---|

**a** Each side adds up to 390. Write the numbers in the circles. Which numbers are in the corners? _____ (Start: try 110 and 120 as two corners. What goes between them?)

**b** Each side adds up to 420. Write the numbers in the circles. Which numbers are in the corners? _____

## No.6

**Mei thinks of two numbers. They add up to 700. One number is 200 more than the other. What are the two numbers? Draw a model.**

**Answer: _____ and _____**

# THINK HARDER

Name _____ Group _____ Date _____

*Draw a model, make a list or make a table. Then write how you know.*

## Stretch

**No.1**

**Find the missing digits: 5A7 + B48 = 905**

**No.2**

**Find the missing digits: 7A1 – B58 = 263**

**No.3**

**What number makes this true? 538 + 199 = 540 + A**

**No.4**

**A shop had 905 eggs. It sold some in the morning and 260 in the afternoon. 345 eggs were left. How many eggs were sold in the morning?**

## Challenge (for strong thinkers)

**No.5**

**Find ALL the ways to fill in the missing digits: AB7 + 4CD = 1000**

**No.6**

**Box A and Box B hold 250 marbles together. Box B and Box C hold 350. Box A and Box C hold 400. How many marbles are in each box?**

**No.7**

**Use the digit cards 1, 2, 3, 4, 5 and 6 once each to make two 3-digit numbers. Make the difference as SMALL as possible.**

**No.8**

**Open investigation: Odd or even?. Ali says: "When I add two odd numbers, I always get an even number." (a) Test Ali's idea with 5 pairs of odd 3-digit numbers (for example 345 + 271). (b) Now test odd + even, and even + even. What do you notice? (c) Does the same happen when you take away? Explain WHY, using pairs of counters or a drawing.**

**What was hard?** No marks for this page. Tick, then write one line.

This lesson felt: ☐ easy ☐ OK ☐ tricky

The hardest part was .....

I got unstuck by: ☐ tried a digit and checked it ☐ worked from the ones column ☐ drew it ☐ asked my partner

Next time I will try .....

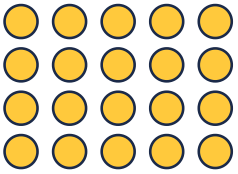
**FAST TRACK.** Both Warm-up check answers right AND you can say why? Go to THINK HARDER and the Open investigation while the group does this page. Rejoin for check-together and talk time.

# LEGS, WHEELS AND ARRAYS

Name _____ Group _____ Date _____

## No.1

**Mei puts counters in an array.**

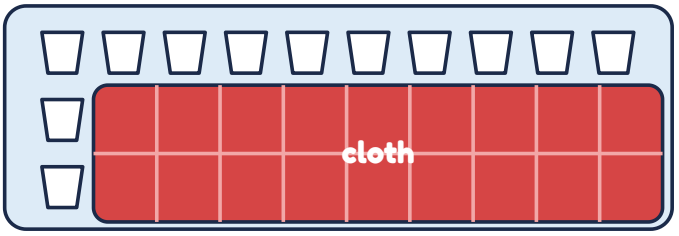


a Two multiplication sentences: _____ × _____ = _____ _____ × _____ = _____

b Two division sentences: _____ ÷ _____ = _____ _____ ÷ _____ = _____

## No.2

**A tray has the same number of cups in every row. A cloth covers some of the cups.**




a How many cups altogether? _____ (b) How many under the cloth? _____

c How did you work it out without lifting the cloth?

## No.3

**A pen has hens and goats. A hen has 2 legs and a goat has 4 legs. There are 7 animals and 20 legs altogether. Start with all hens. Then swap one hen for a goat each time.**

| | | | | | |
|---|---|---|---|---|---|
| **Hens** | 7 | | | | |
| **Goats** | 0 | | | | |
| **Legs** | 14 | | | | |

a How many hens and how many goats? _____ hens _____ goats

## No.4

**Ravi has 23 stickers. He shares them equally among 4 friends, giving each friend as many as he can.**

a How many does each friend get? _____ (b) How many are left over? _____

c What is the smallest number of extra stickers Ravi needs so that none are left over?

_____

## No.5

**28 children go on a trip. Each van holds 5 children.**

a How many vans are needed? _____ (b) How many empty seats? _____

## No.6

**Mr Tan puts 20 chairs in equal rows. There must be at least 2 rows and at least 2 chairs in each row. Find ALL the ways. Start with 2 rows. Stop when the number of rows is bigger than the number of chairs in each row.**

| Rows | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| Chairs in each row | | | | | | | | | |
| Does it work? | | | | | | | | | |

## No.7

**A shop has some bicycles (2 wheels) and some tricycles (3 wheels), at least one of each. There are 17 wheels altogether. Use counters or the table to find ALL the ways. Start with 1 tricycle.**



bicycle: 2 wheels



tricycle: 3 wheels

| Tricycles | 1 | | | | |
|---|---|---|---|---|---|
| Bicycles | | | | | |
| Wheels | | | | | |

a How do you know you have found them all?

.....

# THINK HARDER

Name _____ Group _____ Date _____

*Draw a model, make a list or make a table. Then write how you know.*

## Stretch

**No.1**

**There are 9 hens and goats on a farm. They have 24 legs altogether. How many hens and how many goats are there?**

**No.2**

**Ali shares 34 marbles equally among 4 friends, giving as many as he can. How many does each friend get? How many are left over?**

**No.3**

**An array of stickers has 4 rows. Ben counts 12 stickers in the first 3 rows. How many stickers are in the whole array?**

**No.4**

**27 children go on a boat ride. Each boat holds 4 children. How many boats are needed? How many empty seats are there?**

## Challenge (for strong thinkers)

**No.5**

A toy shop has tricycles (3 wheels) and toy cars (4 wheels), at least one of each. There are 30 wheels altogether. Find all the ways.

**No.6**

Mr Lim has between 20 and 30 sweets. When he puts them in groups of 4, 3 are left over. When he puts them in groups of 5, 2 are left over. How many sweets does he have?

**No.7**

A pen has hens and goats. There are 2 more hens than goats. There are 22 legs altogether. How many hens and how many goats?

**No.8**

Open investigation: Where tables meet. (a) List the numbers up to 50 that are in BOTH the 2 times table and the 5 times table. What do you notice about them? (b) Now find the numbers up to 50 that are in both the 3 and 5 tables, and in both the 4 and 5 tables. (c) Is every number in the 10 times table in both the 2 and 5 tables? Is every number in both the 2 and 5 tables in the 10 table? Explain.

**What was hard?** No marks for this page. Tick, then write one line.

This lesson felt: ☐  easy ☐  OK ☐  tricky

The hardest part was .....

I got unstuck by: ☐ made a table ☐ used counters ☐ drew arrays ☐ asked my partner

Next time I will try .....

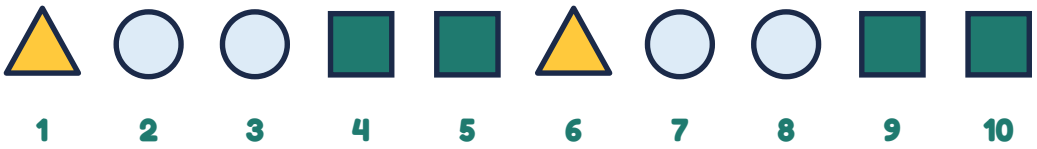
**FAST TRACK.** Both Warm-up check answers right AND you can say why? Go to THINK HARDER and the Open investigation while the group does this page. Rejoin for check-together and talk time.

# PATTERN MACHINES

Name _____ Group _____ Date _____

## No.1

**This pattern keeps going in the same way. Here are places 1 to 10.**



1 2 3 4 5 6 7 8 9 10

place

a What shape is in place 20? _____ (b) In place 23? _____

c What shape is in place 41? _____ How do you know?

d How many circles are there in the first 30 shapes? _____

## No.2

**Ali writes the letters M, A, T, H again and again.**




a What is the 18th letter? _____ (b) Where is the 10th letter H? place _____

c Ali writes 30 letters. How many Ts does he write? _____ Show how you know.

## No.3

**Siti's bead string goes 2 red, 3 yellow, 2 red, 3 yellow ... She uses 12 red beads. The string ends with 3 yellow beads. Tip: draw the string in groups, or thread beads on a pencil with counters.**




a How many yellow beads? _____ (b) How many beads altogether? _____

## No.4

**Each flower has 5 petals.**



**a** Fill in the table.

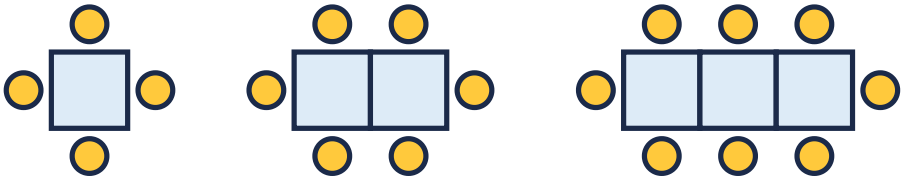
| **Flowers** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Petals** | 5 | | | | | |

**b** Petals on 10 flowers? _____ (c) 35 petals make how many flowers? _____

**d** Mei has 23 petals. How many flowers? _____ Petals left over? _____

## No.5

**Square tables are pushed together in a row. One child sits at each open side.**



**1 table** **2 tables** **3 tables**

**a** Fill in the table.

| **Tables** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Children** | 4 | 6 | | | | |

**b** Children at 10 tables? _____ (c) Tables for 20 children? _____

**d** Why does the number of children go up by 2 each time?

.....

## No.6

**Ali counts in 3s: 3, 6, 9 ... Mei counts in 4s: 4, 8, 12 ... Which numbers from 1 to 36 do they BOTH say? Make two lists.**

**a** Ali: 3, 6, 9, _____

**b** Mei: 4, 8, 12, _____

**c** Both say: _____ What do you notice? _____

# THINK HARDER

Name _____ Group _____ Date _____

*Draw a model, make a list or make a table. Then write how you know.*

## Stretch

**No.1**

**A pattern goes A, B, B, C, A, B, B, C ... What is the 25th letter?**

**No.2**

**Beads go red, yellow, yellow, red, yellow, yellow ... How many yellow beads are in the first 18 beads?**

**No.3**

**Houses are made of sticks in a row. 1 house uses 5 sticks, 2 houses use 9 sticks, 3 houses use 13 sticks. How many sticks are needed for 6 houses?**

**No.4**

**Ali counts in 4s from 4. Mei counts in 10s from 10. Which numbers up to 50 do they both say?**

## Challenge (for strong thinkers)

**No.5**

Ali counts in 3s from 3. Mei counts in 4s from 4. They both say 12. Predict the next number they will BOTH say. Then list every number up to 100 that they both say. What do you notice? Why?

**No.6**

Siti builds towers of cubes. Tower 1 has 2 cubes, tower 2 has 5 cubes, tower 3 has 8 cubes. The towers keep growing in the same way. Which tower has 29 cubes? Is there a tower with exactly 50 cubes? With exactly 100 cubes? Explain how you know.

**No.7**

Frog A starts at 0 and jumps 4 each time: 4, 8, 12 ... Frog B starts at 2 and jumps 3 each time: 5, 8, 11 ... Which numbers up to 30 do both frogs land on?

**No.8**

Open investigation: Three counting patterns. Pattern A: 1, 4, 7, 10 ... (add 3 each time). Pattern B: 2, 5, 8, 11 ... Pattern C: 3, 6, 9, 12 ... (a) Write each pattern up to 30. (b) Pick any number from 1 to 30. Which pattern is it in? Can a number be in two patterns? Can a number be in none? (c) Why does this happen? (d) What if the patterns added 4 each time: how many patterns would you need to catch every number?

**What was hard?** No marks for this page. Tick, then write one line.

This lesson felt: ☐  easy ☐  OK ☐  tricky

The hardest part was .....

I got unstuck by: ☐ counted in groups ☐ wrote the pattern out ☐ made a table ☐ asked my partner

Next time I will try .....

**FAST TRACK.** Both Warm-up check answers right AND you can say why? Go to THINK HARDER and the Open investigation while the group does this page. Rejoin for check-together and talk time.

# MYSTERY NUMBER MACHINES

Name _____ Group _____ Date _____

## No.1

**Mystery machines. Find the number that went IN and write it in the yellow box. Undo the LAST step first, then check forwards.**

**a** I multiply by 2, then add 7. I get 27.

| | | | | | | |
|---|---|---|---|---|---|---|
| **?** | → | × 2 | → | + 7 | → | **27** |

**b** I take away 8, then multiply by 5. I get 40.

| | | | | | | |
|---|---|---|---|---|---|---|
| **?** | → | − 8 | → | × 5 | → | **40** |

**c** I divide by 3, then add 12. I get 20.

| | | | | | | |
|---|---|---|---|---|---|---|
| **?** | → | ÷ 3 | → | + 12 | → | **20** |

**d** I multiply by 4, then take away 15. I get 21.

| | | | | | | |
|---|---|---|---|---|---|---|
| **?** | → | × 4 | → | − 15 | → | **21** |

## No.2

**Two-step number machines.**

**a** This machine does × 3, then + 2. Fill in the table.

| **IN** | 4 | 10 | | |
|---|---|---|---|---|
| **OUT** | | | 17 | 29 |

**b** This machine multiplies by a number, then adds a number. Find what it does. (Try × 2, × 3, × 4 … with IN 2, then test IN 5.)

| **IN** | 2 | 5 | 10 | |
|---|---|---|---|---|
| **OUT** | 11 | 20 | | 26 |

**c** It does × _____ , then + _____ . IN 10 gives OUT _____ . OUT 26 comes from IN _____ .

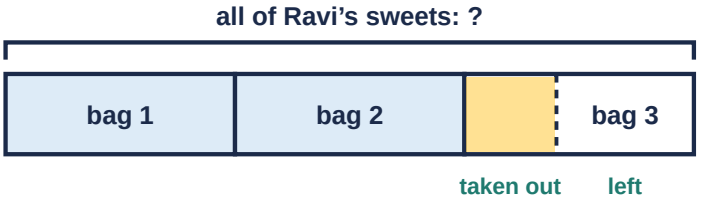
**No.3**

**Siti had some stickers. She gave 24 stickers to her brother. Then her mother gave her 15 more. Now she has 50 stickers. How many stickers did Siti have at first? Draw before-and-after bars.**

**Answer:** _____

**No.4**

**Ravi put all his sweets into 3 bags, with the same number in each bag. He took 4 sweets out of bag 3. Now bag 3 has 6 sweets. How many sweets did Ravi have altogether?**




**Answer:** _____

**No.5**

**Mei and Ali both put 5 into a machine.**

**a** Mei's machine does + 4, then × 2. What comes out? _____

**b** Ali's machine does × 2, then + 4. What comes out? _____

**c** Do the two machines always give the same answer? _____

**d** What must Ali put IN to get the same OUT as Mei? _____

# THINK HARDER

Name _____ Group _____ Date _____

*Draw a model, make a list or make a table. Then write how you know.*

## Stretch

**No.1**

**A machine does × 4, then + 13. The OUT is 45. What went IN?**

**No.2**

**Ben had some money. He spent $15 on a book. Then his dad gave him $10. Now he has $42. How much money did Ben have at first?**

**No.3**

**Siti had some beads. She gave half of them to Mei. Then she bought 12 more. Now she has 30 beads. How many beads did Siti have at first?**

**No.4**

**Ali, Ben and Siti have 40 marbles altogether. Ben has 5 more marbles than Ali. Siti has 8 more marbles than Ali. How many marbles does each child have? Draw a bar for each child.**

## Challenge (for strong thinkers)

**No.5**

Siti had some sweets. She gave Ben 1 more than half of them. Now she has 7 sweets left. How many sweets did she have at first?

**No.6**

A machine does × 2, then – 1. Mei puts a number through the machine 3 times in a row (each OUT goes back in). The final OUT is 17. What number did she start with?

**No.7**

Three boxes have 30 pens altogether. 5 pens are moved from box A to box B. Then 3 pens are moved from box B to box C. Now all three boxes have the same number of pens. How many pens were in each box at first?

**No.8**

Open investigation: A magic trick. Think of a number. Add 5. Double it. Take away 10. Halve it. (a) Try 5 different starting numbers. What do you notice about your final answer? (b) Is it ALWAYS true? Try a big number and a small number, then say why. (c) Make your own trick that always gives you back the number you started with. Test it on 3 numbers.

**What was hard?** No marks for this page. Tick, then write one line.

This lesson felt: ☐  easy ☐  OK ☐  tricky

The hardest part was .....

I got unstuck by: ☐ drew the machine ☐ worked backwards ☐ tried a number and checked it ☐ asked my partner

Next time I will try .....

**FAST TRACK.** Both Warm-up check answers right AND you can say why? Go to THINK HARDER and the Open investigation while the group does this page. Rejoin for check-together and talk time.

# MARKET DAY

Name _____ Group _____ Date _____

Coins: 5¢, 10¢, 20¢, 50¢, $1. Notes: $2, $5, $10, $50, $100. (There is no $20 note.) 100¢ = $1.

## No.1

**Pay exactly. Use the FEWEST notes and coins you can.**

| Item | Price | Notes and coins I use | Pieces |
|---|---|---|---|
| Curry puff | $1.65 | | |
| Notebook | $4.30 | | |
| Umbrella | $18.75 | | |
| Bicycle | $76 | | |

## No.2

**Find EVERY way to pay exactly $8 using $5 notes, $2 notes and $1 coins. Make a list in order: start with the most $5 notes.**

| Way | Number of $5 | Number of $2 | Number of $1 | Pieces altogether |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | | | |
| D | | | | |
| E | | | | |
| F | | | | |
| G | | | | |
| H | | | | |

a How many ways are there? _____

b Two ways use exactly 4 pieces of money. What are they?

**Answer:** _____

## No.3

**Change.**

**a** Mei buys an eraser for 65¢. She pays with a $1 coin. How much change does she get?

_____

**b** Ali buys a game for $26. He pays with a $50 note. How much change does he get? Which notes give this change with the fewest notes?

**Answer:** _____

## No.4

**Siti has $20 to spend.**




**a** Can Siti buy the bag and the cap? Explain.

**b** Siti buys two different things and gets exactly $3 change. Find ALL the pairs she could buy.

## No.5

**Work backwards.**

**a** Mei spent 45¢ on a sweet and 30¢ on a pencil. She has 25¢ left. How much money did she have at first? Write it in dollars and cents. _____

**b** Ben spent half of his money on a kite. Then he spent $4 on a ball. He had $6 left. How much money did Ben have at first? Tip: draw his money as a bar cut in half.

_____

## No.6

**Stickers cost 20¢ each. Erasers cost 50¢ each. Siti spends exactly $2. How many stickers and erasers could she buy? Find ALL the ways. (She might buy only one kind.)**

# THINK HARDER

Name _____ Group _____ Date _____

*Draw a model, make a list or make a table. Then write how you know.*

## Stretch

**No.1**

**How many ways can you pay exactly $10 using $5 notes, $2 notes and $1 coins?**

**No.2**

**Ali has 4 coins. They are worth $1.25 altogether. Find all the sets of 4 coins he could have.**

**No.3**

**How many different ways can you pay exactly 40¢ using 5¢, 10¢ and 20¢ coins? Make a list in order, starting with the most 20¢ coins.**

**No.4**

**Pens cost $3 each and books cost $7 each. Ben buys 6 things and spends exactly $30. How many pens and how many books does he buy? Make a table.**

## Challenge (for strong thinkers)

**No.5**

Siti had some money. She spent half of it on a book. Then she spent half of what was left on a pen. She had $5 left. How much money did she have at first?

**No.6**

Sweets cost 5¢ each and lollipops cost 20¢ each. Ben buys 8 things altogether and spends exactly $1. How many sweets and how many lollipops does he buy?

**No.7**

Mei has one $10 note, one $5 note, one $2 note and one $1 coin. How many different amounts can she pay exactly, using one or more of them?

**No.8**

Open investigation: Most coins. Use only 5¢, 10¢, 20¢ and 50¢ coins. (a) Pay 35¢ using the fewest coins. How many coins is that? (b) Try the other amounts from 5¢ to 95¢ (in steps of 5¢). Make a table of how many coins each needs. (c) Which amount needs the MOST coins, even when you use the fewest possible? Is there more than one? Why do they need so many?

**What was hard?** No marks for this page. Tick, then write one line.

This lesson felt: ☐  easy ☐  OK ☐  tricky

The hardest part was .....

I got unstuck by: ☐ made a list in order ☐ used play coins ☐ made a table ☐ asked my partner

Next time I will try .....

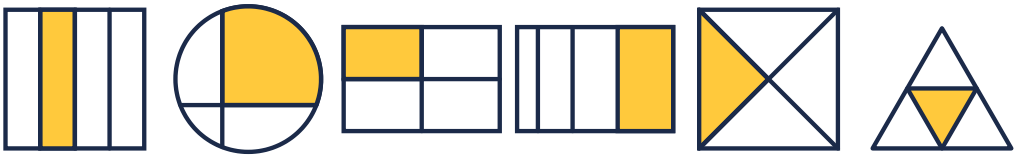
**FAST TRACK.** Both Warm-up check answers right AND you can say why? Go to THINK HARDER and the Open investigation while the group does this page. Rejoin for check-together and talk time.

# FAIR SHARES

Name _____ Group _____ Date _____

## No.1

**Fair shares? Each shape has 4 parts and 1 part is shaded. Circle the shapes that are cut into quarters (4 EQUAL parts).**



A B C D E F

a Choose one shape that is NOT cut into quarters. Explain how you know.

.....

## No.2

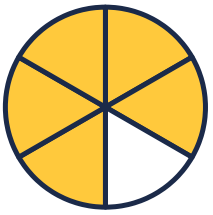
**What fraction is shaded? What fraction is NOT shaded? Write them in the boxes.**

a



shaded ☐ not shaded ☐

b



shaded ☐ not shaded ☐

c



shaded ☐ not shaded ☐

## No.3

**Compare the fractions. Draw bars the same length if it helps.**

a Circle the bigger fraction in each pair: $\frac{1}{3}$ or $\frac{1}{8}$ $\frac{5}{9}$ or $\frac{7}{9}$

b Smallest first: $\frac{1}{5}$ $\frac{1}{2}$ $\frac{1}{10}$ $\frac{1}{4}$ _____ _____ _____ _____

c Mei says: "$\frac{1}{6}$ is bigger than $\frac{1}{4}$, because 6 is bigger than 4." Is she right? Explain.

.....

## No.4

**Add or take away. Colour the bars to help you. Write each answer in the box.**

**a** 

$\frac{2}{5} + \frac{1}{5} = \square$

**b** 

$\frac{3}{8} + \frac{4}{8} = \square$

**c** 

$\frac{4}{12} + \frac{5}{12} = \square$

**d** 

$\frac{1}{6} + \square = 1 \text{ whole}$

**e** 

$\frac{7}{10} - \frac{3}{10} = \square$

## No.5

**Mum cuts a cake into 8 equal slices. Ali eats 2 slices. Mei eats 3 slices.**

**Cake** 

**a** What fraction of the cake do they eat altogether? _____

**b** What fraction of the cake is left? _____

**c** Who eats more? How much more? _____

## No.6

**Ravi and Siti each have a pizza. The pizzas are the same size. Ravi cuts his into 4 equal slices and eats 1 slice. Siti cuts hers into 8 equal slices and eats 3 slices. Who eats more pizza? Hint: fold or line up the bars; one quarter covers two eighths. Colour the bars to show how you know.**

**Ravi** 

**Siti** 

**Who eats more?** _____

.....

# THINK HARDER

Name _____ Group _____ Date _____

*Draw a model, make a list or make a table. Then write how you know.*

## Stretch

**No.1**

**Write these in order, smallest first: 1/3, 1/12, 1/2, 1/5.**

**No.2**

**A ribbon is cut into 9 equal pieces. Mei uses 4 pieces and Siti uses 3 pieces. What fraction of the ribbon is left?**

**No.3**

**2/7 + ? = 1 whole**

**No.4**

**Ali says: "Half of my pizza is bigger than half of your pizza." Can he be right? Explain.**

## Challenge (for strong thinkers)

**No.5**

**A cake is cut into 12 equal pieces. Mei gets 4/12. Ali and Ben share the rest, but Ali gets 2/12 more than Ben. What fraction does Ben get?**

**No.6**

**A paper strip is folded in half, then in half again, then in half again. When you open it, how many equal parts are there? What fraction is each part?**

**No.7**

**Siti eats 1/2 of a pizza. Then Ravi eats 1/2 of what is left. What fraction of the whole pizza is left now?**

**No.8**

**Open investigation: Fraction strips. Use your fraction strips (halves, thirds, quarters, sixths and eighths). (a) Cover a HALF strip exactly with pieces of ONE kind only. Which kinds work? How many pieces of each? (b) Do the same for a THIRD strip and a QUARTER strip. (c) Can thirds ever cover a half exactly? Why not?**

**What was hard?** No marks for this page. Tick, then write one line.

This lesson felt: ☐ easy ☐ OK ☐ tricky

The hardest part was .....

I got unstuck by: ☐ folded paper strips ☐ used the fraction strips ☐ drew bars ☐ asked my partner

Next time I will try .....

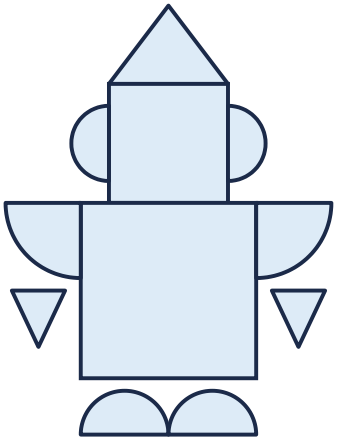
**FAST TRACK.** Both Warm-up check answers right AND you can say why? Go to THINK HARDER and the Open investigation while the group does this page. Rejoin for check-together and talk time.

# SHAPE BUILDERS

Name _____ Group _____ Date _____

## No.1

**The robot is made of pieces. Count each kind of piece and write how many in the boxes. Tick each piece as you count it.**



- Square ☐
- Triangle ☐
- Semicircle ☐
- Quarter circle ☐

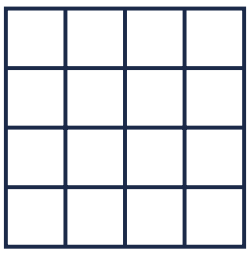
a Which kind is there the most of? _____ b Pieces altogether? _____

c Mei has 3 semicircles and 6 quarter circles, all cut from same-size circles. How many whole circles can she make? Show how.

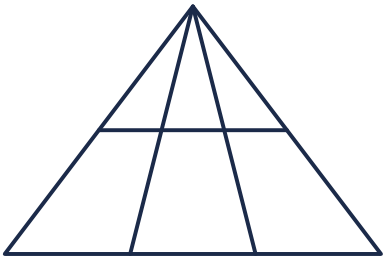
## No.2

**Count every shape. Work in order.**

a



b

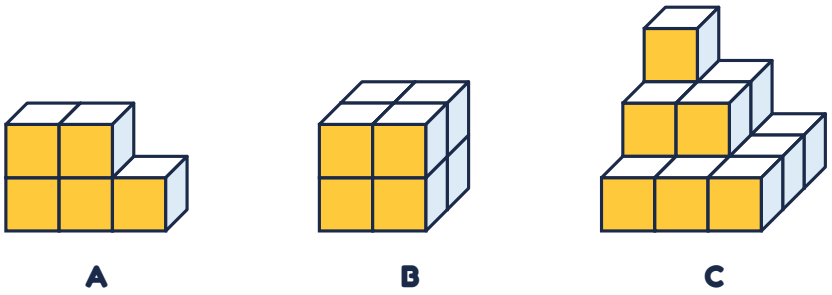


| a 1 by 1 | 2 by 2 | 3 by 3 | 4 by 4 | Squares altogether |
|---|---|---|---|---|
| | | | | |

b Triangles on the bottom line: _____ on the line across: _____ altogether: _____

## No.3

**Count the cubes. Every cube is on the table or sits on top of another cube. Stack A is only 1 cube deep.**



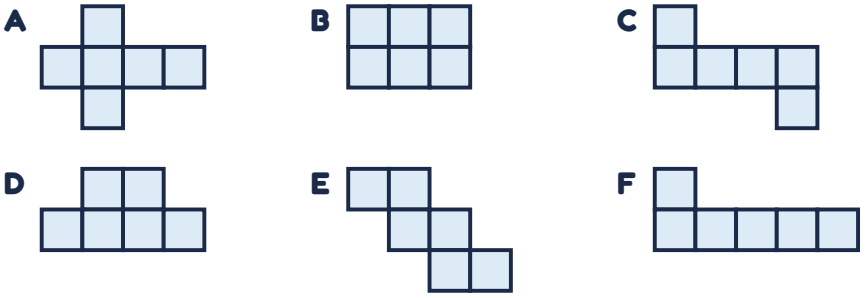
A B C

**a** A: _____ cubes B: _____ cubes C: _____ cubes

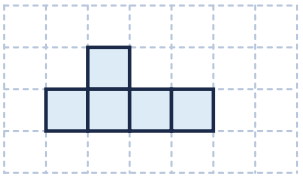
**b** Ravi wants to turn stack C into a big cube 3 cubes long, 3 cubes wide and 3 cubes high. How many more cubes does he need? _____

## No.4

**Cut out nets A to F from the cut-out page and try folding each one. Then circle the letters of the nets that fold up into a cube.**




**b** Add ONE more square to this shape so that it folds into a cube. Draw it on the grid.



# THINK HARDER

Name _____ Group _____ Date _____

*Draw a model, make a list or make a table. Then write how you know.*

## Stretch

**No.1**

**A stack of cubes has a bottom layer 4 cubes long and 3 cubes deep, a middle layer 3 cubes long and 2 cubes deep, and 1 cube on top. How many cubes are in the stack? Draw the layers.**

**No.2**

**How many squares are in a grid 3 squares long and 2 squares high?**

**No.3**

**A stack has 2 layers. Each layer is 3 cubes long and 2 cubes wide. How many cubes?**

**No.4**

**Six squares are joined in a straight line. Can they fold into a cube? Explain.**

## Challenge (for strong thinkers)

**No.5**

A pyramid of cubes has a top layer of 1 cube, then a layer 2 by 2, then a layer 3 by 3, then a bottom layer 4 by 4. How many cubes are there?

**No.6**

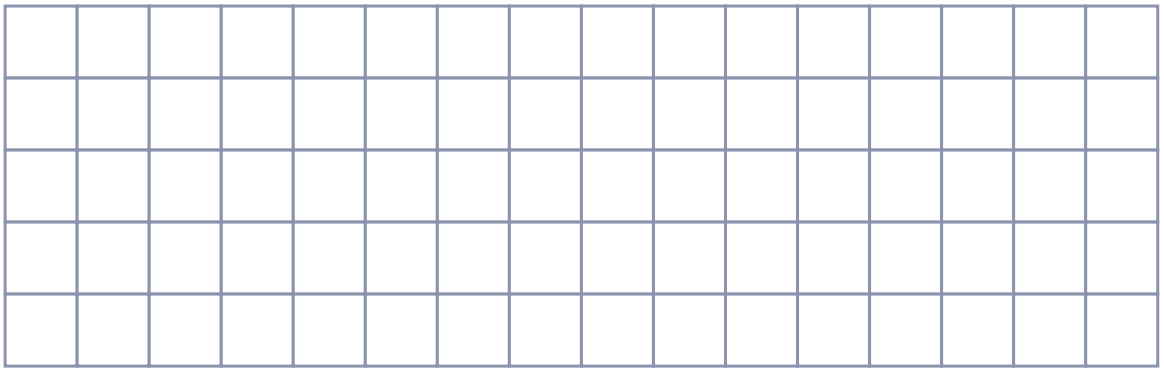
How many squares can you find in a grid 5 squares long and 5 squares high?

**No.7**

On a dice, opposite faces add up to 7. You can see 3 faces showing 1, 2 and 3. What numbers are on the 3 faces you cannot see? What do they add up to?

**No.8**

**Open investigation: Squares joined edge to edge. Use square tiles (or coloured squares of paper) and the grid below. Squares must touch along a whole side. A shape that is turned round or flipped over counts as the SAME shape. (a) Find ALL the different shapes you can make with 3 squares. (b) Now find ALL the different shapes you can make with 4 squares. Draw each one on the grid. (c) How do you know you have found them all?**



**What was hard?** No marks for this page. Tick, then write one line.

This lesson felt: ☐ 🙂 easy ☐ 😐 OK ☐ ☹ tricky

The hardest part was .....

I got unstuck by: ☐ used square tiles ☐ drew it on the grid ☐ made a list ☐ asked my partner

Next time I will try .....

**FAST TRACK.** Both Warm-up check answers right AND you can say why? Go to THINK HARDER and the Open investigation while the group does this page. Rejoin for check-together and talk time.

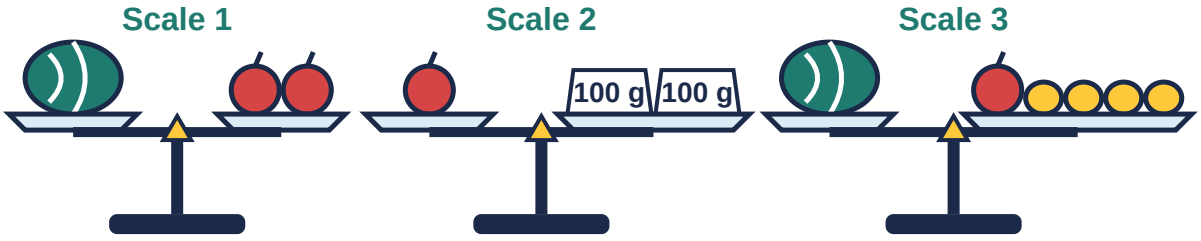
# MEASURE MASTERS

**Name** _____ **Group** _____ **Date** _____

A level balance means both sides are equally heavy. Show how you know for every answer.

## No.1

**Each scale is level.**




**a** How heavy is 1 apple? _____ g

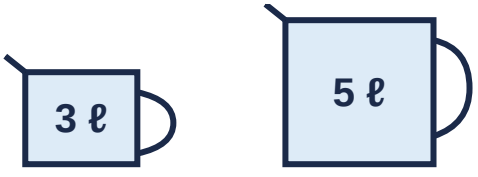
**b** How heavy is the melon? _____ g

**c** How heavy is 1 lemon? Show how you know. Tip: on Scale 3, melon = apple + 4 lemons.

**d** How many lemons balance the melon? _____ lemons

## No.2

**You have a 3 ℓ jug and a 5 ℓ jug. You can fill a jug from the tap, empty a jug into the sink, or pour from one jug into the other until one jug is full or the other is empty.**




No marks on the jugs!

**a** How can you get exactly 2 ℓ? Write the steps.

**b** How can you get exactly 1 ℓ? Write the steps.

## No.3

**Next puzzle. Start where Q2 (b) ended: 1 ℓ in the 3 ℓ jug and 5 ℓ in the 5 ℓ jug. Now get exactly 4 ℓ in the 5 ℓ jug. Write each pour in the table. You may not need every row.**

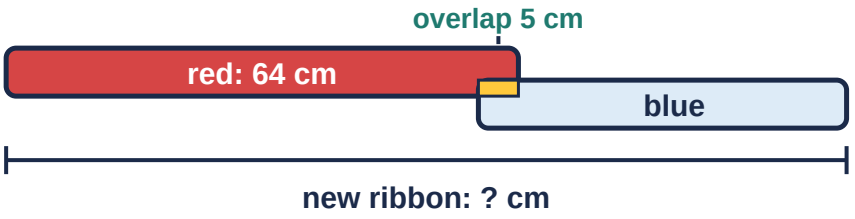
| Step | What I do | 3 ℓ jug | 5 ℓ jug |
|---|---|---|---|
| Start | (where Q2 b ended) | 1 ℓ | 5 ℓ |
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |

## No.4

**Siti has a red ribbon 64 cm long. Her blue ribbon is 18 cm shorter than the red ribbon.**

**a** How long is the blue ribbon? _____ cm

**b** She glues them end to end with a 5 cm overlap. The new ribbon is _____ cm long.




**c** Ben cuts a 20 m rope into 4 equal pieces. Each piece: _____ m Cuts: _____

## No.5

**Read the clocks. Hint: on clock B the hour hand is almost on the 5. Is it 4 o'clock or 5 o'clock?**




Ali is eating breakfast.




Ali is at swimming after school.

short thick hand = hour long red hand = minute

**a** Write each time with a.m. or p.m. A: _____ B: _____

**b** Swimming starts at the time on clock B and lasts 35 minutes. It ends at _____

**c** Mei's family leaves home at 10:45 a.m. They reach Grandma's flat at 12:10 p.m. How long is the trip? _____

# THINK HARDER

Name _____ Group _____ Date _____

*Draw a model, make a list or make a table. Then write how you know.*

## Stretch

**No.1**

**3 apples balance 1 pomelo. The pomelo and 2 apples together balance a 500 g weight. How heavy is one apple? How heavy is the pomelo?**

**No.2**

**You have a 2 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 3 ℓ? List each pour.**

**No.3**

**Mei's piano lesson ends at 4:15 p.m. It lasted 45 minutes. What time did it start?**

**No.4**

**Buses to the MRT station leave every 15 minutes. The first bus leaves at 6:50 a.m. Ravi reaches the bus stop at 7:40 a.m. How long does he wait? The ride takes 25 minutes. When does he reach the MRT station?**

## Challenge (for strong thinkers)

**No.5**

You have a 3 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 5 ℓ? List each pour.

**No.6**

Scale 1: 2 cubes and 1 ball balance 2 balls. Scale 2: 1 ball and 1 cube balance a 300 g weight. How heavy is one cube? How heavy is one ball?

**No.7**

Mei has 3 lessons in a row. Each lesson is 35 minutes long, with a 10-minute break between lessons. The first lesson starts at 8:15 a.m. What time does the last lesson end?

**No.8**

Open investigation: Weights puzzle. You have three different weights. Each weight is 1 g, 2 g, 3 g, 4 g or 5 g. You put some of them on one pan to make a mass. (a) With weights of 1 g, 2 g and 4 g, which masses from 1 g to 7 g can you make? (b) Find ALL the sets of three different weights from 1 g to 5 g that can make every mass from 1 g to 6 g. (c) What do all the good sets have in common?

**What was hard?** No marks for this page. Tick, then write one line.

This lesson felt: ☐  easy ☐  OK ☐  tricky

The hardest part was .....

I got unstuck by: ☐ made a table ☐ used cubes or coins as weights ☐ listed in order ☐ asked my partner

Next time I will try .....

**FAST TRACK.** Both Warm-up check answers right AND you can say why? Go to THINK HARDER and the Open investigation while the group does this page. Rejoin for check-together and talk time.

# CHALLENGE DAY

Name _____ Group _____ Date _____

## No.1

**The picture graph shows the ice-creams a stall sold.**

| | |
|---|---|
| Monday |  |
| Tuesday |  |
| Wednesday |  |
| Thursday | Draw the pictures for Thursday here. |

 = 5 ice-creams

**a** How many ice-creams were sold on Tuesday? _____

**b** 90 ice-creams were sold in the 4 days. Draw Thursday's pictures on the graph.

**c** Each ice-cream costs $2. How much did the stall get on Wednesday? _____

## No.2

**Use the digit cards 3, 0 and 8 once each to make a 3-digit number.**

**a** List ALL the numbers in order. _____

**b** Greatest: _____ Smallest: _____

**c** Which of them are between 300 and 810? _____

**No.3**

**Ben has 6 coins. They are all 50¢ and 20¢ coins. Together they are worth $1.80. How many of each coin does he have? Make a table.**

| Number of 50¢ | Number of 20¢ | Total |
|---|---|---|
| 0 | 6 | $1.20 |
| | | |
| | | |
| | | |

Ben has _____ 50¢ coins and _____ 20¢ coins.

## No.4

**Ali, Mei, Siti and Ben each go to school a different way: bus, MRT, car or walking. Put a tick where it fits and a cross where it cannot.**

Clue 1: Siti lives next to the school, so she walks.

Clue 2: Ben does not go by bus or by car.

Clue 3: Mei does not go by car.

| | Bus | MRT | Car | Walk |
|---|---|---|---|---|
| **Ali** | | | | |
| **Mei** | | | | |
| **Siti** | | | | |
| **Ben** | | | | |

Ali: _____ Mei: _____ Siti: _____ Ben: _____

## No.5

**A cake is cut into 10 equal pieces. Ali eats 3/10 of the cake. Mei eats 2 more pieces than Ali. Ben eats the rest.**



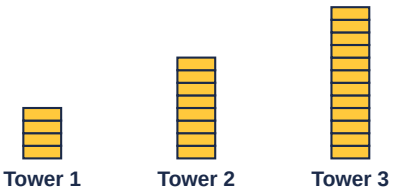
a What fraction of the cake does Mei eat? _____

b What fraction of the cake does Ben eat? _____

c Order the children, the one who eats the most first. _____

## No.6

**Ben builds towers with cubes. Tower 1 has 4 cubes, Tower 2 has 8 cubes and Tower 3 has 12 cubes. Keep the pattern going.**




| Tower | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Cubes** | 4 | 8 | 12 | | | |

a How many cubes are in Tower 8? _____

b Which tower has 36 cubes? Tower _____

# THINK HARDER

Name _____ Group _____ Date _____

*Draw a model, make a list or make a table. Then write how you know.*

## Stretch

**No.1**

**30 children chose a fruit. In the picture graph each picture stands for 5 children. Apple has 2 pictures and banana has 3 pictures. The mango row is missing. How many pictures should the mango row have?**

**No.2**

**How many 3-digit numbers have digits that add up to 3? List them all.**

**No.3**

**A bus had some people on it. At the first stop half of the people got off. At the second stop 6 people got on. Now there are 14 people on the bus. How many people were on the bus at the start? Draw a bar to start.**

**No.4**

**5, 9, 13, 17, ... What is the 10th number in this pattern?**

## Challenge (for strong thinkers)

**No.5**

A picture graph has lost its key. Football has 6 pictures and swimming has 4 pictures. 10 more children chose football than swimming. How many children chose swimming? Football?

**No.6**

Ali, Ben, Mei and Siti each have one coin: 10¢, 20¢, 50¢ or $1. Mei has more money than Ali but less than Ben. Ben does not have the $1 coin. Who has which coin?

**No.7**

At a hawker centre, chicken rice costs $4 a plate and noodles cost $3 a bowl. Ravi's family buys 7 dishes for $25. How many plates of chicken rice do they buy? What is the smallest total 7 dishes can cost? The biggest? Can 7 dishes cost $30? Can they cost EVERY whole-dollar amount in between?

**No.8**

Open investigation: Staircase. Mei climbs a staircase. She can take 1 step or 2 steps at a time. For 3 steps there are 3 ways: 1+1+1, 1+2 and 2+1. (a) Find ALL the ways for 4 steps. (b) Find ALL the ways for 5 steps. (c) Fill in a table for 1, 2, 3, 4 and 5 steps. What pattern do you see? Can you guess the number of ways for 6 steps?

**What was hard?** No marks for this page. Tick, then write one line.

This lesson felt: ☐  easy ☐  OK ☐  tricky

The hardest part was .....

I got unstuck by: ☐ listed in order ☐ made a table ☐ drew it ☐ asked my partner

Next time I will try .....

# Cut-outs

For the P2 program. Cut along the dashed lines and keep the pieces in one envelope.

## Place-value cards (Lessons 1, 2 and 10)

| | | |
|---|---|---|
| 100 | 10 | 1 |
| 200 | 20 | 2 |
| 300 | 30 | 3 |
| 400 | 40 | 4 |
| 500 | 50 | 5 |
| 600 | 60 | 6 |
| 700 | 70 | 7 |
| 800 | 80 | 8 |
| 900 | 90 | 9 |

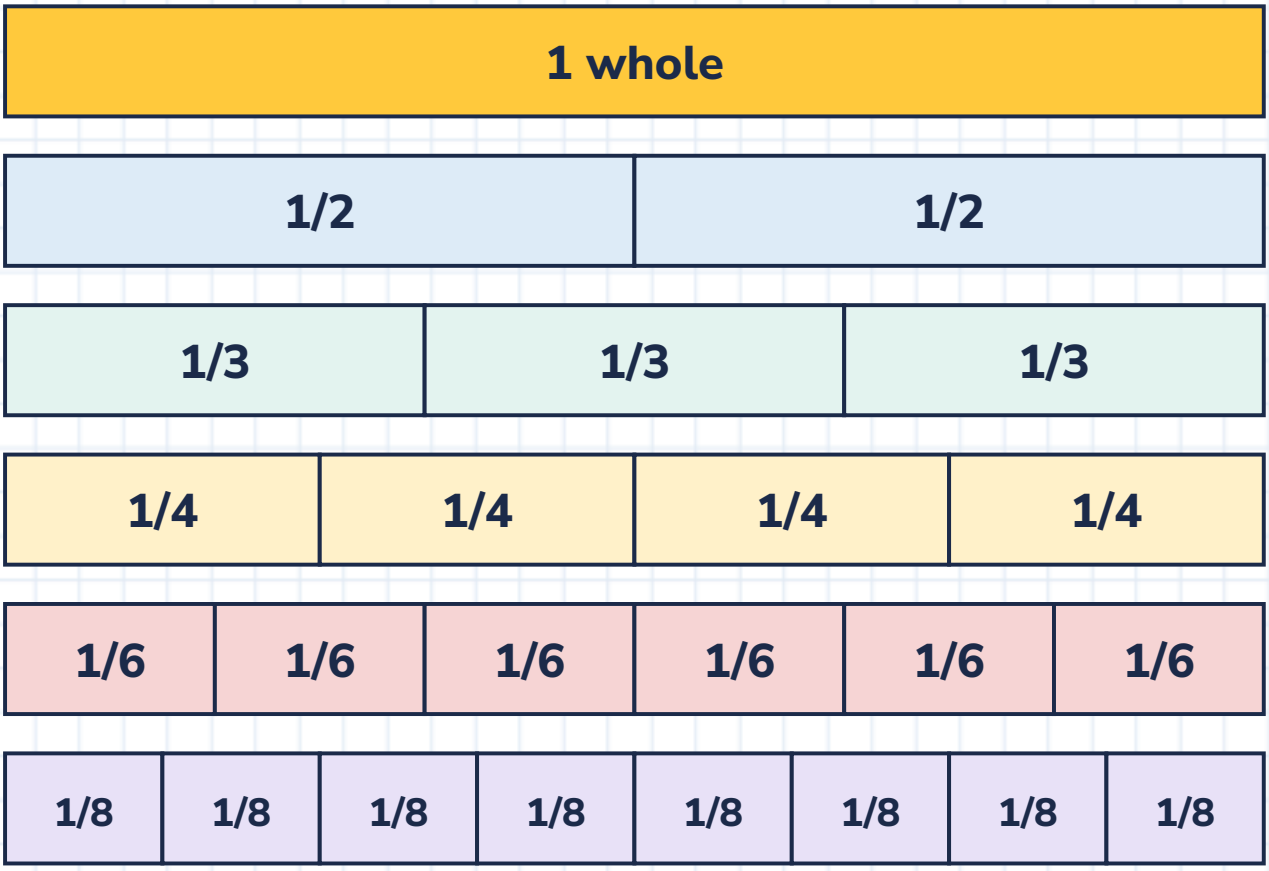
Lay the cards on top of each other, right edges together: 300 + 40 + 7 shows 347.

# Digit cards

Two sets of 0 to 9 (Lessons 1 and 10).

| | | | | |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 | 9 |
| 0 | 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 | 9 |

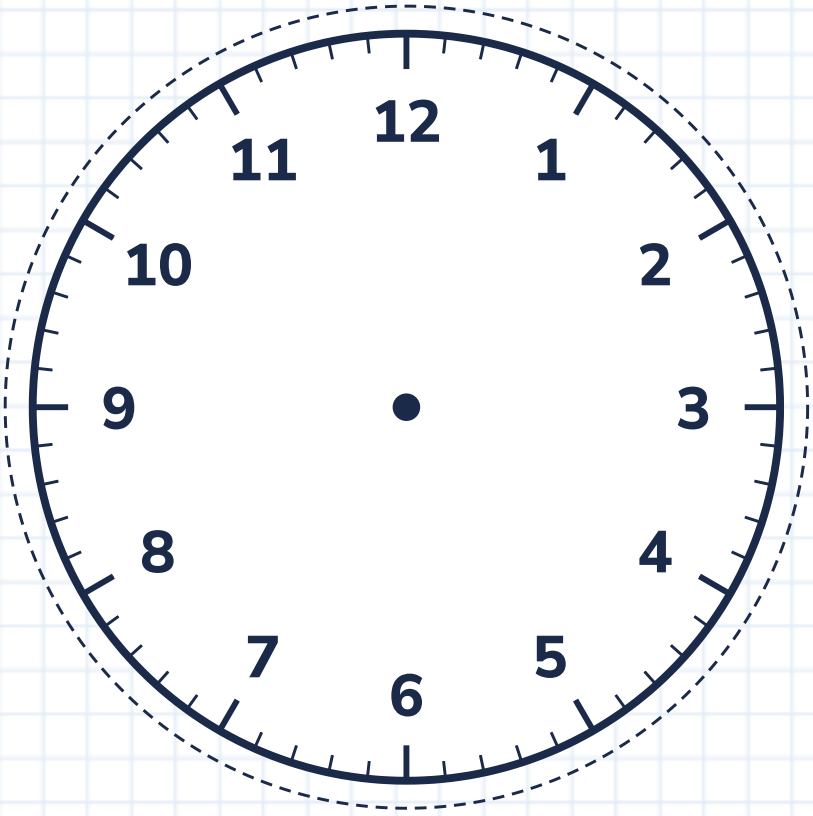
## Fraction strips (Lesson 7)

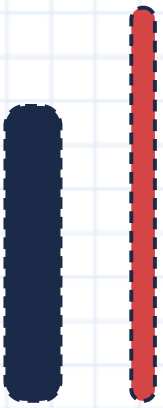



Cut along each strip, then cut the parts apart. Compare: which is bigger, 1/3 or 1/4?

# Paper clock

For Lesson 9 (time).



**hour hand (short, thick)**

**minute hand (long, thin)**

Cut out the clock and the hands. Pin them at the centre with a split pin or a short pencil.

## Paper strips of numbered squares (Lesson 9: ribbons and lengths)

### Strip A

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

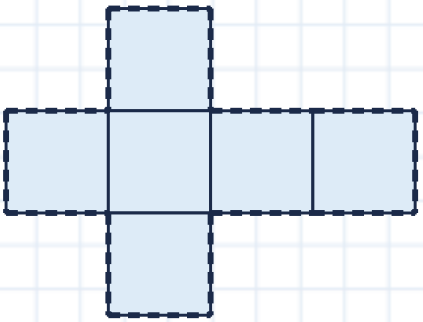
### Strip B

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

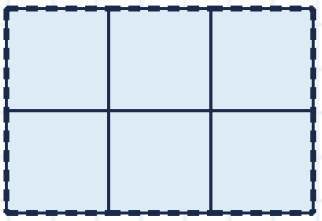
# Cube nets

For Lesson 8 (Shape Builders). Cut out each net and try folding it. Which ones make a cube?

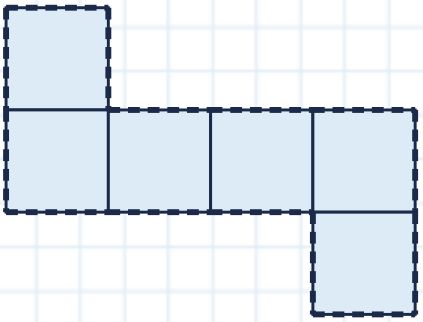
## Net A



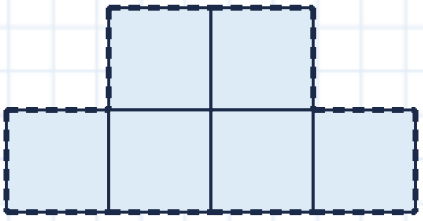
## Net B



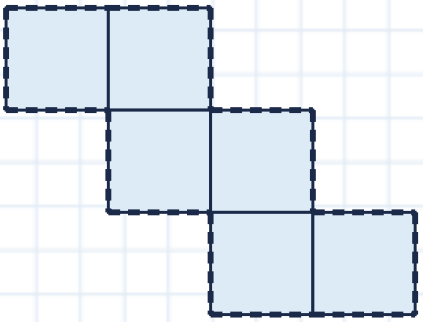
## Net C



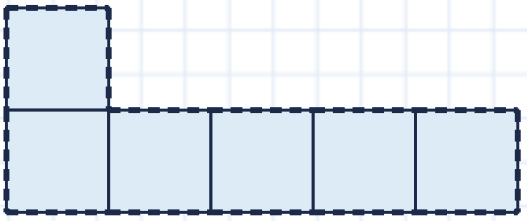
## Net D



## Net E



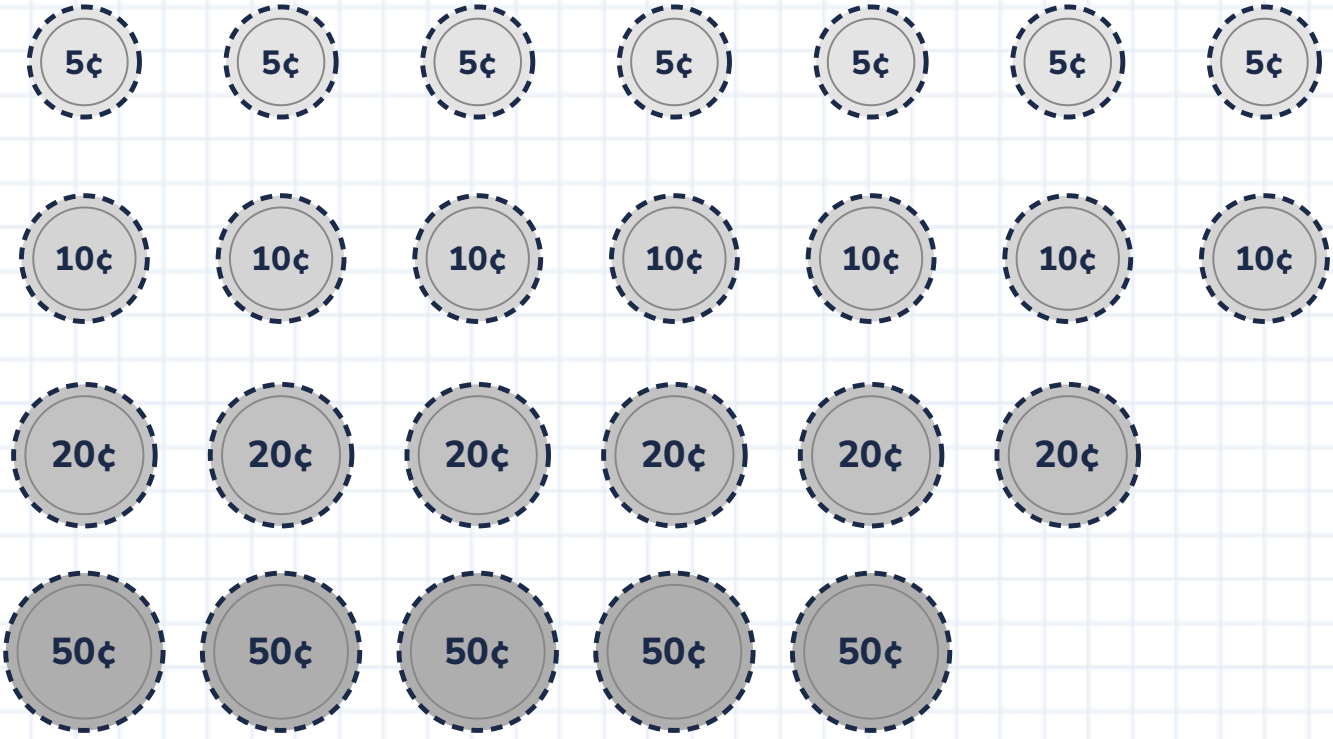
## Net F



Cut along the dashed lines. Fold along the solid lines. Which nets fold up into a cube?

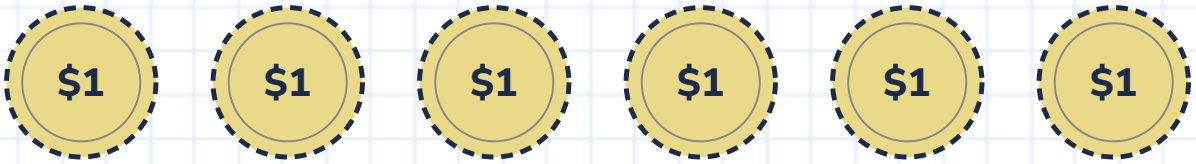
# Play coins

For Lesson 6 (Market Day). Seven 5¢, seven 10¢, six 20¢ and five 50¢. Keep spare coins in the envelope.



# Play notes and $1 coins

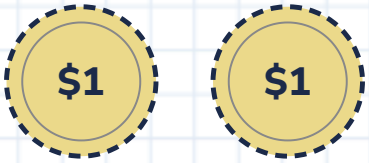
For Lesson 6 (Market Day). Eight $1 coins (six here and two spare below), four $2, four $5, four $10 and two $50 notes. Keep spare coins in the envelope.







## Two spare $1 coins



# CERTIFICATE OF COMPLETION

## Math Inquirers Holiday Program

Primary 2 Holiday Program

This certifies that

.....

completed all 10 lessons, solved puzzles, spotted patterns and explained their thinking.

## Notice it. Test it. Explain why.

Tutor: _____ Date: _____